

Visual Atlas

Every diagnostic figure, one per panel, each with its takeaway

Companion archive: VISUAL_ATLAS_FIGURES.zip — every figure as a standalone PNG with a matching .txt giving its description, interpretation and takeaway

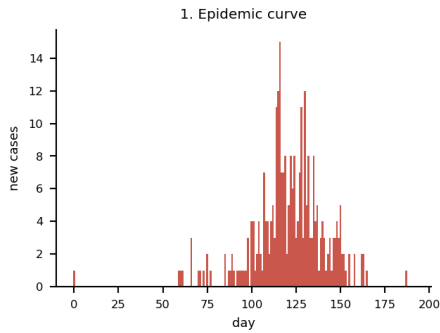
HOW TO READ THIS DOCUMENT — the bold line under each figure is the takeaway. Read only those and you have the whole argument. Section 6 covers the study design and the undocumented-pathogen question.

Contents

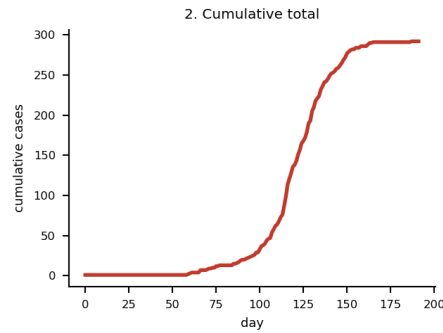
Section	Outbreak	Figures	What it establishes
1	Ebola, Kikwit 1995	18	clean incidence series; all concerns resolved
2	Measles, Hagelloch 1861	18	complete cohort; clean timing, clustered bursts
3	Influenza, England 1978	19	method benchmark; three limits that cannot be fixed
4	Norovirus, Derbyshire 2001	14	individual-level records; most precise arm
5	Across outbreaks	15	comparability, design, and the unknown-pathogen question

1. Ebola — Kikwit, DR Congo, 1995

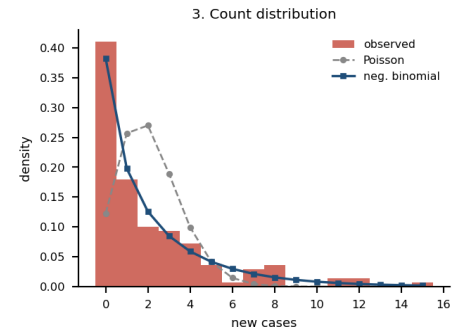
292 cases over six months · daily new-case counts · RECON / Khan et al. (1999)



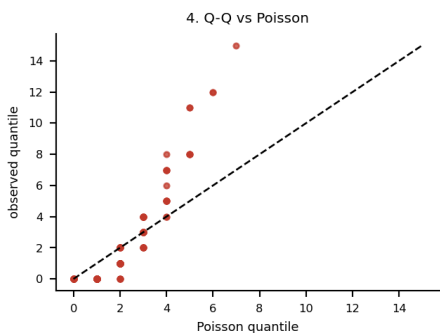
One rise and one fall: a single wave that exhausted its susceptibles and stopped.



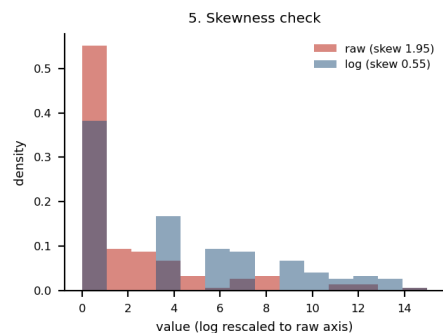
The S-shape confirms saturation: growth slows because susceptibles run out, not by chance.



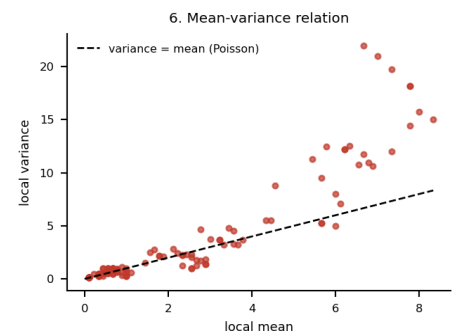
The observed spread is wider than Poisson allows, so a negative binomial is required.



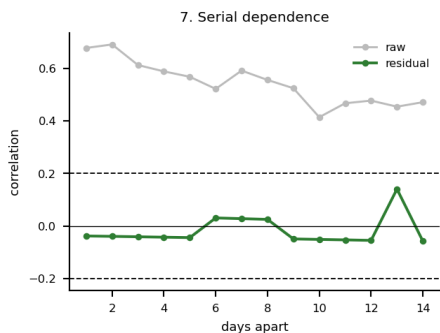
Points curving above the line in the upper tail means extreme days are more common than Poisson predicts.



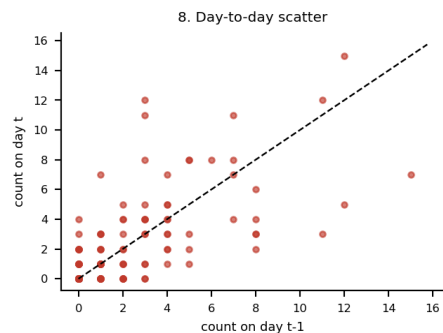
Skewness 1.95 on the raw scale; the log scale is far more symmetric, confirming multiplicative rather than additive variation.



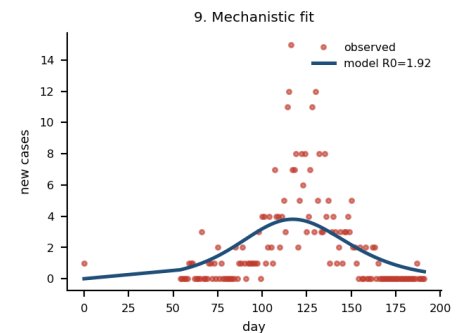
Variance grows faster than the mean, the signature of clustered spread rather than independent events.



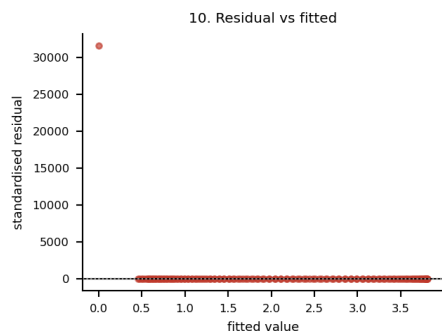
Raw dependence +0.68 falls to -0.04 once the mechanism is fitted - fully absorbed.



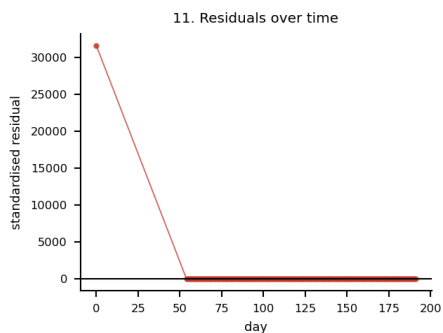
A tight diagonal cloud means today resembles yesterday; the model must explain that, not ignore it.



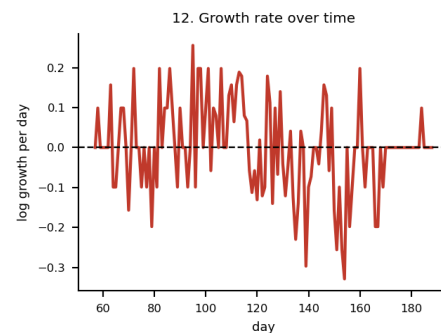
The fitted spread score is 1.92; the curve tracks the observed rise and fall.



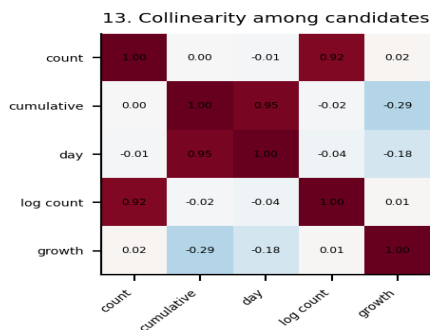
No funnel or curve means the chosen variance structure is appropriate across the whole range.



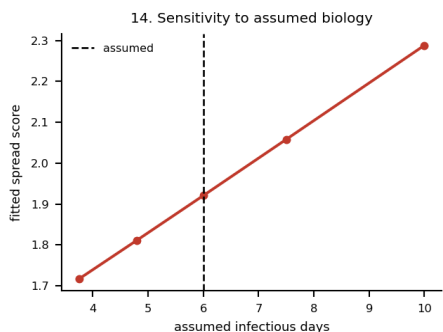
Scatter around zero with no drift means the model is not systematically early or late.



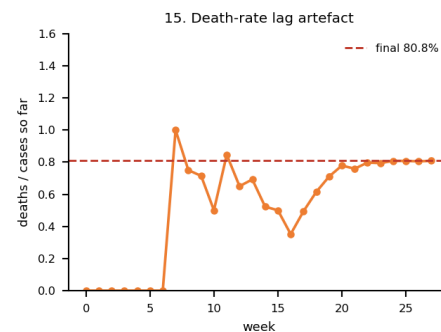
Growth starts positive, crosses zero at the peak, then turns negative - the classic single-wave signature.



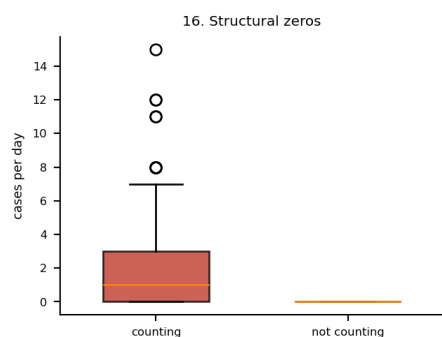
Cumulative totals correlate strongly with each other, so only one time-derived quantity may enter the model.



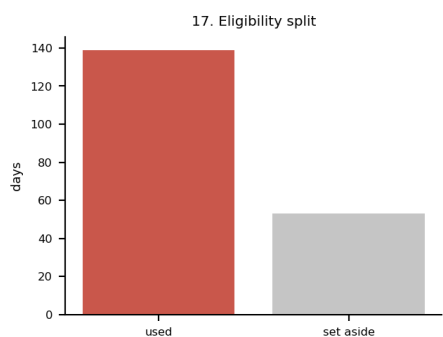
The estimate moves with the assumed biology, so the assumed values are reported alongside the result rather than hidden.



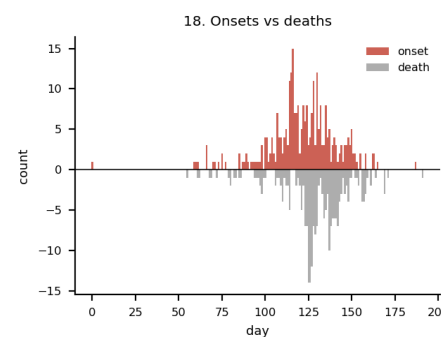
Unstable for ten weeks purely because deaths trail cases; day-level death rate is therefore never a model input.



Near-total separation ($p=5e-13$): non-reporting days carry no information and are excluded rather than counted as zeros.



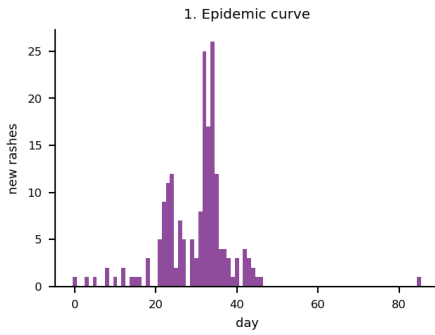
139 of 192 days are usable; the excluded 53 are retained in the file so the decision stays auditable.



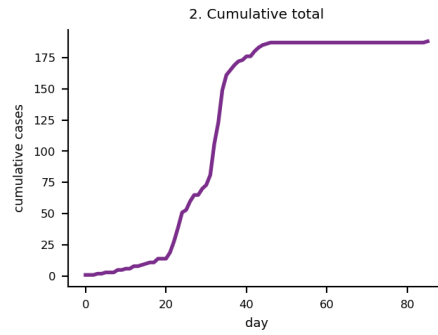
Deaths mirror onsets about two weeks later, which is the lag that makes early death rates meaningless.

2. Measles — Hagelloch, Germany, 1861

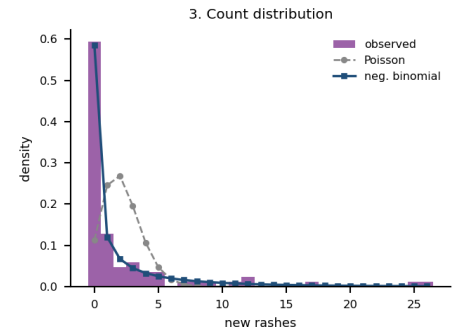
every child in one village · daily new-rash counts · Pfeilsticker (1863); Neal & Roberts (2004)



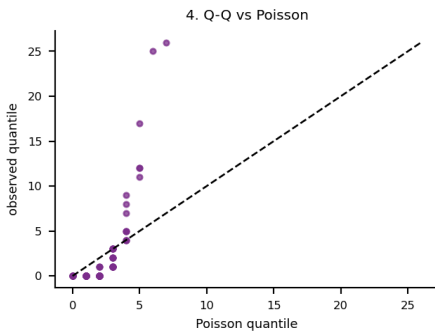
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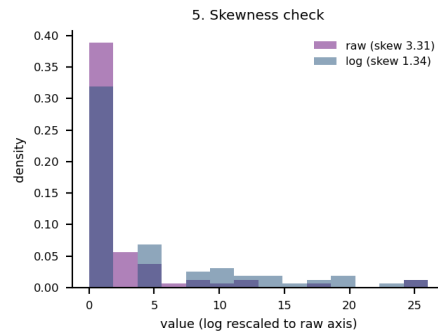
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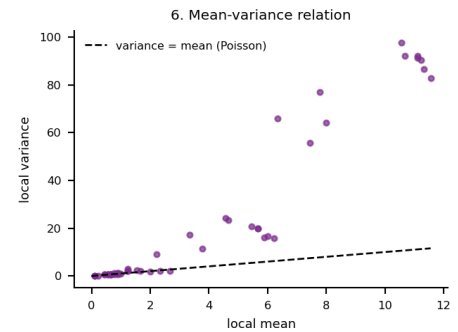
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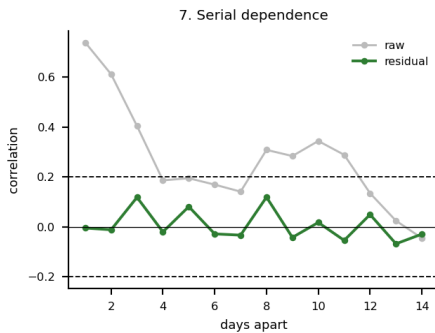
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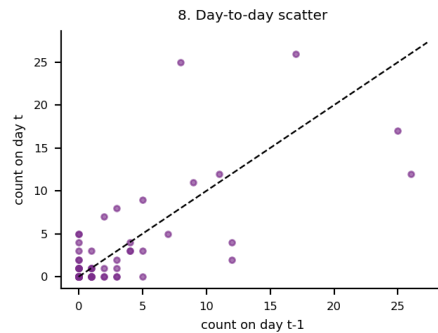
Skewness 3.31 on the raw scale; the log scale is far more symmetric, confirming multiplicative rather than additive variation.



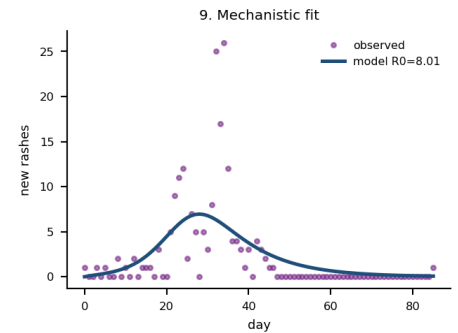
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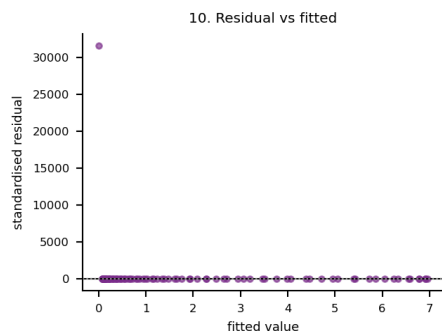
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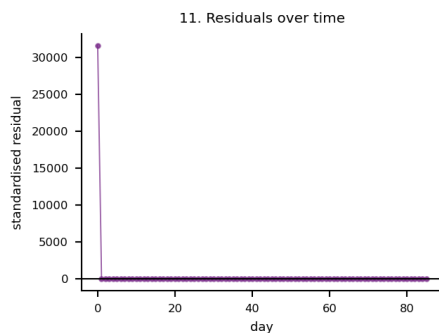
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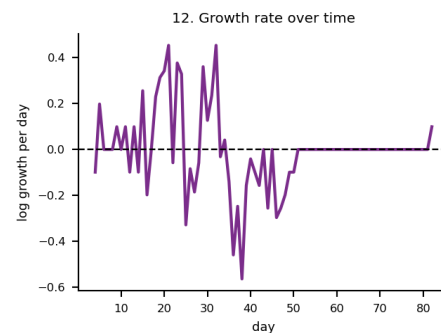
The fitted spread score is 8.01; the curve tracks the observed rise and fall.



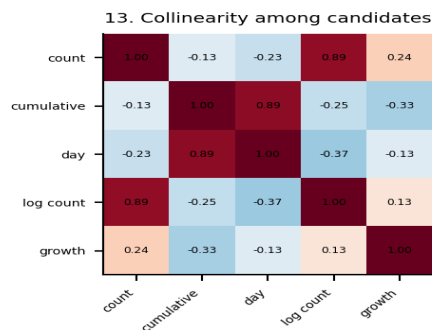
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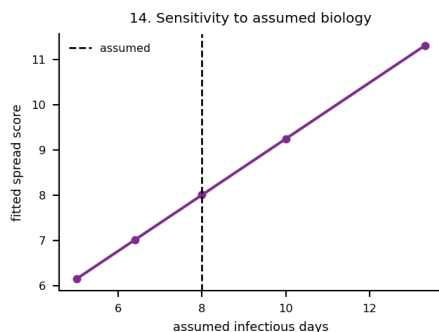
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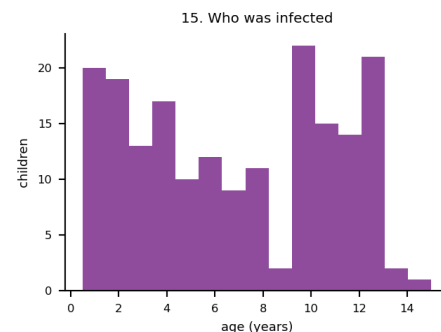
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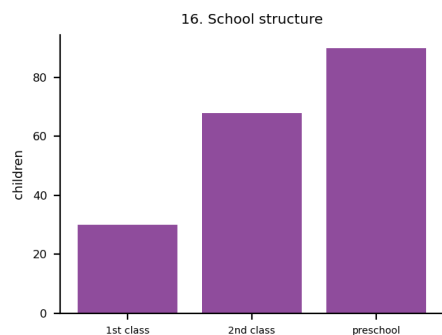
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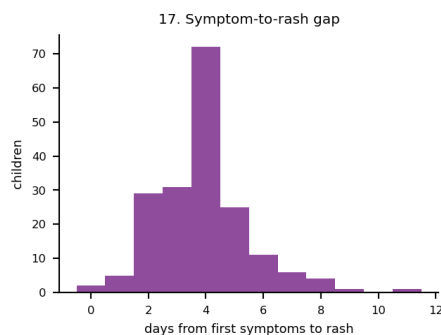
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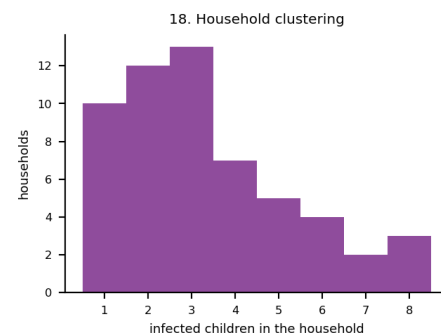
Every age from infant to fifteen was hit, confirming no pre-existing immunity anywhere in the cohort.



Three distinct mixing groups; the mid-epidemic burst is the disease moving through classrooms.



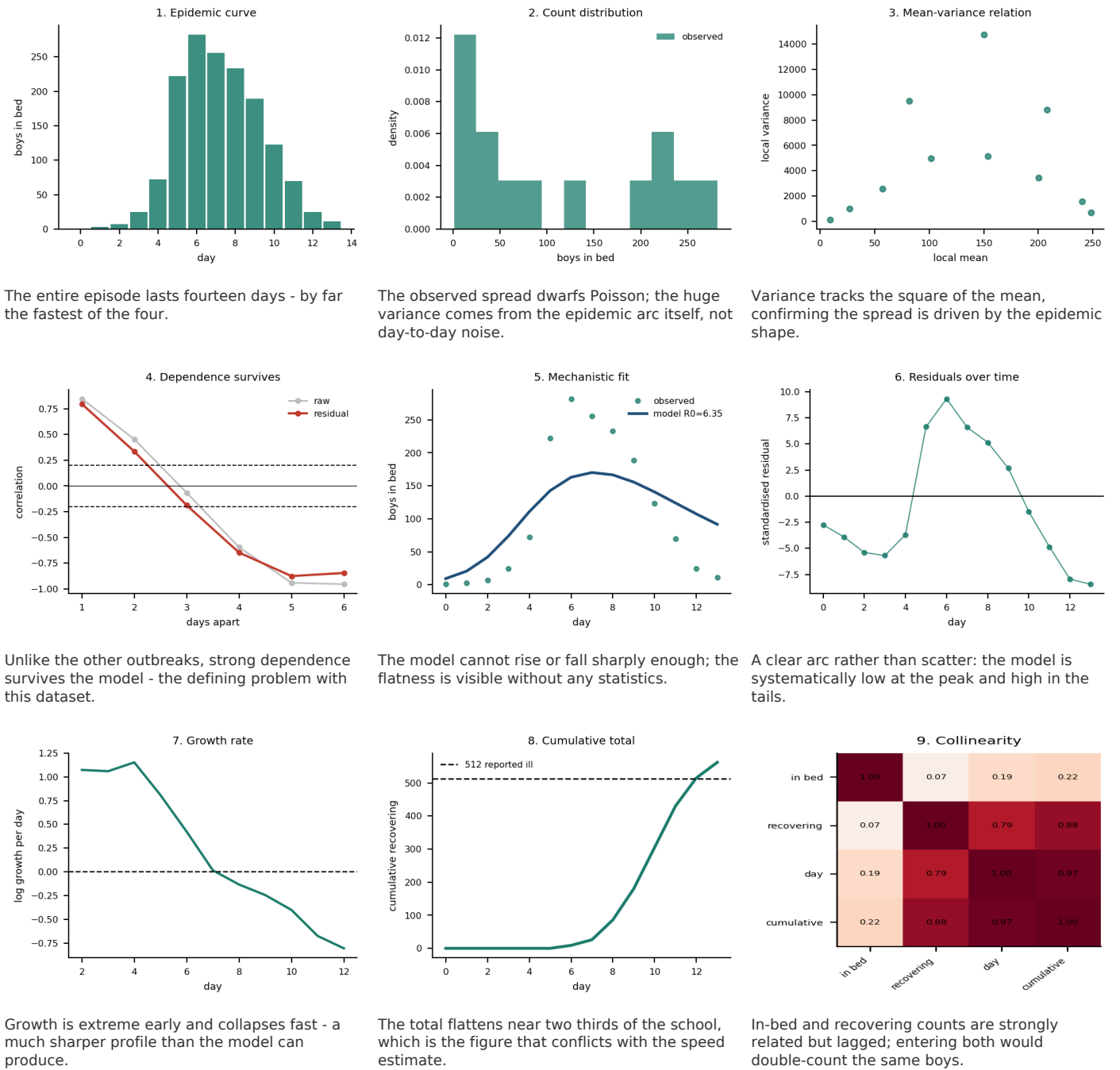
A consistent four-day gap justifies treating rash onset as a clean, uniformly shifted marker of infection.



Several households contributed multiple cases, which is why the real curve is spikier than an even-mixing model draws.

3. Influenza — English boarding school, 1978

CAVEATED ARM · daily boys-in-bed counts · BMJ (1978); Avilov et al. (2024)



4. Dependence survives

Unlike the other outbreaks, strong dependence survives the model - the defining problem with this dataset.

5. Mechanistic fit

The model cannot rise or fall sharply enough; the flatness is visible without any statistics.

6. Residuals over time

A clear arc rather than scatter: the model is systematically low at the peak and high in the tails.

7. Growth rate

Growth is extreme early and collapses fast - a much sharper profile than the model can produce.

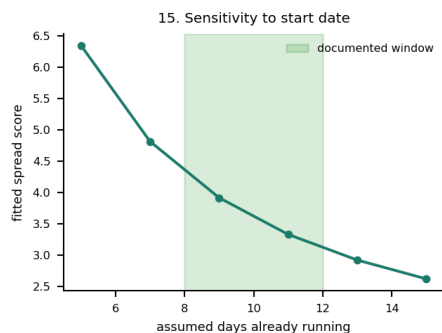
8. Cumulative total

The total flattens near two thirds of the school, which is the figure that conflicts with the speed estimate.

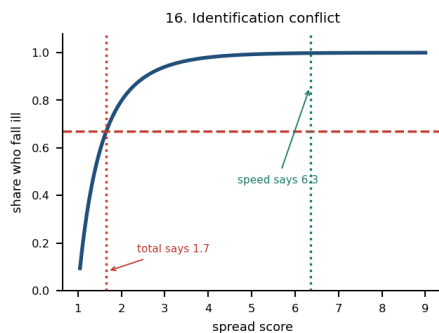
9. Collinearity

in bed	1.00	0.07	0.19	0.22
recovering	0.07	1.00	0.79	0.88
day	0.19	0.79	1.00	0.97
cumulative	0.22	0.88	0.97	1.00

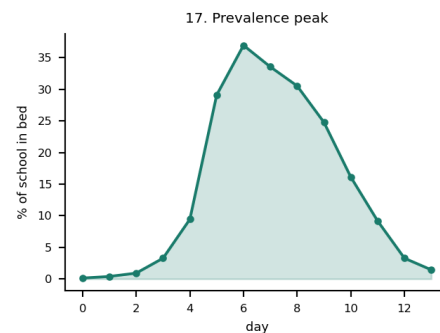
In-bed and recovering counts are strongly related but lagged; entering both would double-count the same boys.



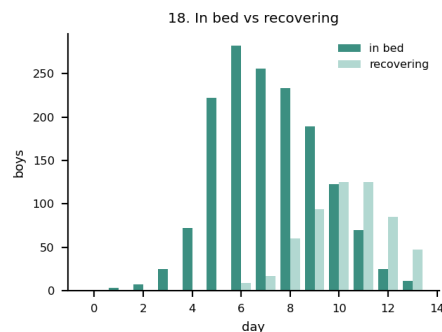
The answer swings with an assumption the records do not pin down, which is the first of three unresolved limits.



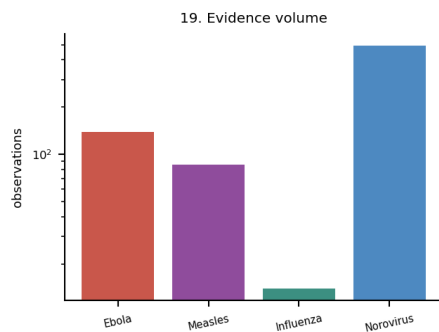
Total infected and epidemic speed imply different answers; no single value satisfies both.



More than a third of the school was bedridden simultaneously - an extreme prevalence peak in a closed population.



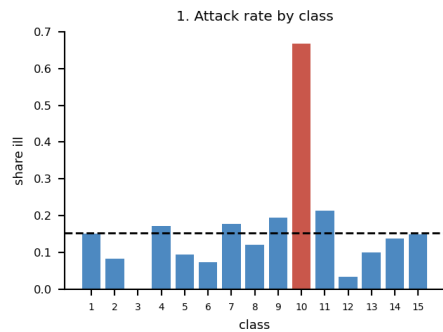
The recovering group lags the in-bed group, confirming in-bed is a holding state, not the infectious state.



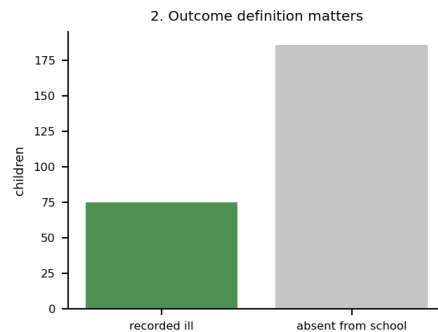
Fourteen daily tallies against 492 individual children: the thinnest evidence base in the study.

4. Norovirus — Derbyshire school, England, 2001

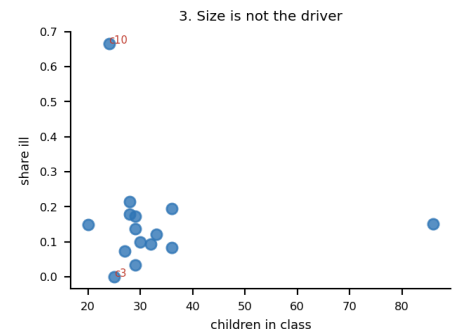
492 children recorded individually · O'Neill & Marks (2005)



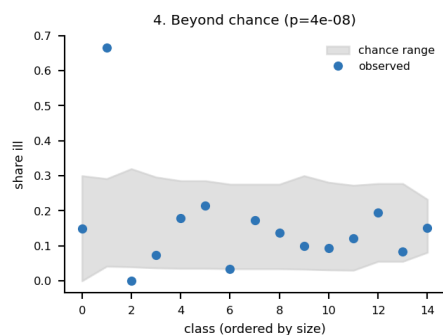
Attack rates run from 0% to 67% between classrooms - clustering, not an even sprinkling.



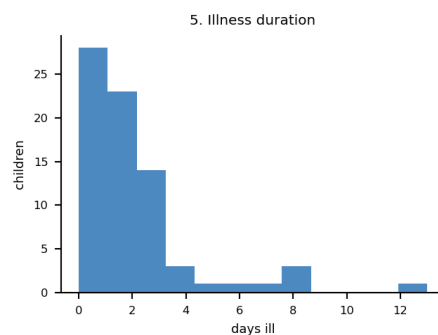
126 children were absent but never recorded ill; using absence would have inflated the outbreak 2.5-fold.



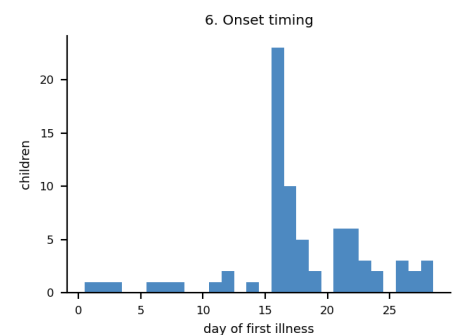
No size effect, so the clustering reflects contact patterns rather than crowding.



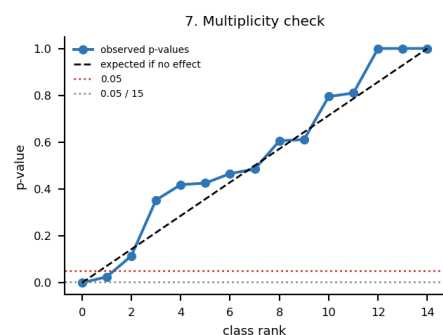
Several classes fall outside the chance band, which is formal evidence that classroom must enter the model.



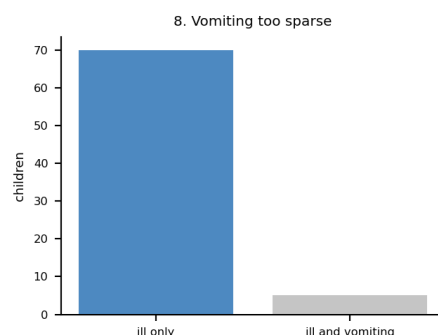
Most recovered within two days: a mild, short, self-limiting infection.



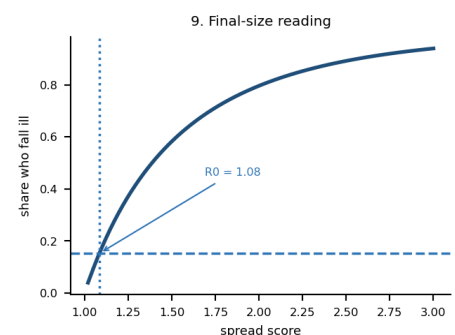
Cases cluster into a short burst, consistent with a common exposure rather than a long chain.



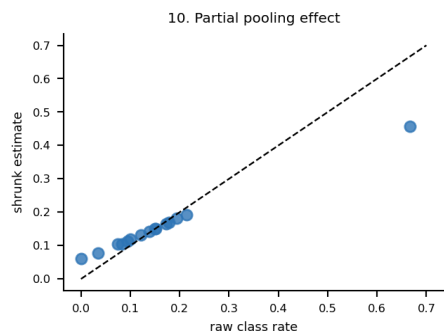
Testing fifteen classrooms separately would produce false positives; one pooled model with class effects avoids that.



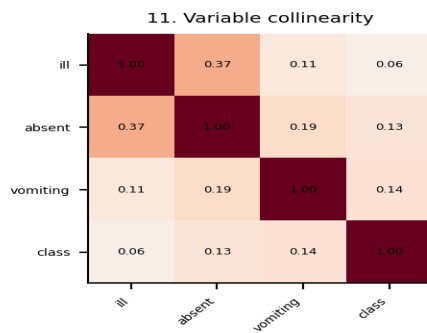
Only 5 children have both recorded - far too few to model, so vomiting is excluded.



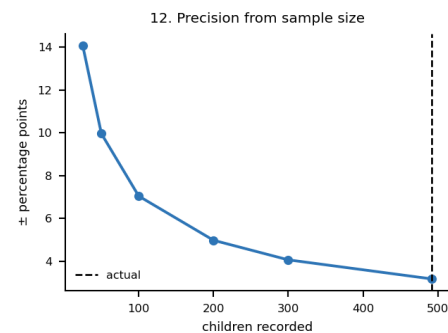
Reading 15% off the curve gives a spread score of 1.08, at the low end of institutional norovirus estimates.



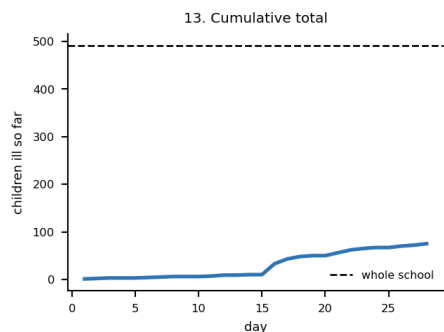
Extreme classrooms are pulled toward the school average rather than deleted, so no class dominates the result.



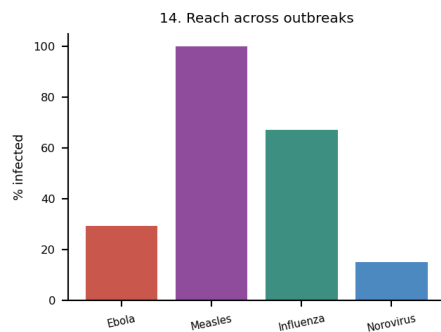
Absence correlates with illness but is not the same thing; entering both would double-count the outcome.



492 individual records is why this is the most precise arm; a daily tally of the same outbreak would be far vaguer.



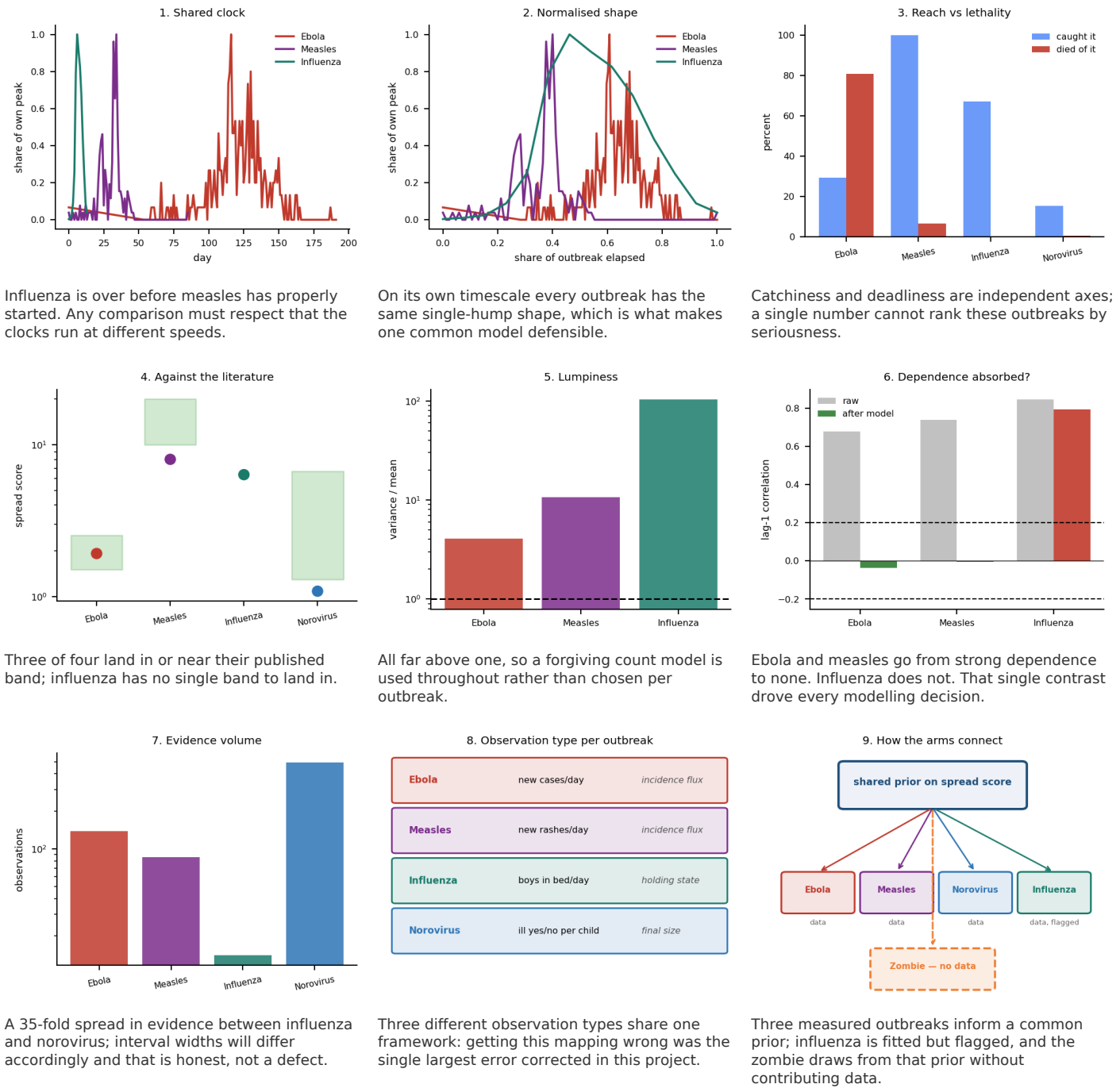
The curve flattens early: the outbreak stopped well short of the whole school, unlike measles.

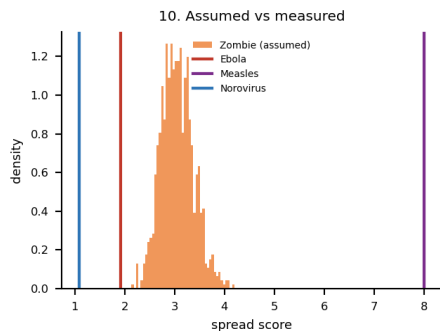


At 15%, norovirus is the mildest of the four by reach - the opposite extreme from measles at 100%.

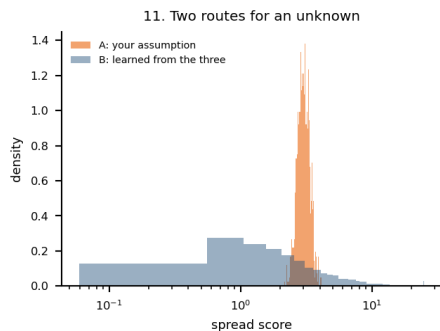
5. Across Outbreaks

Comparability, study design, and the undocumented-pathogen question

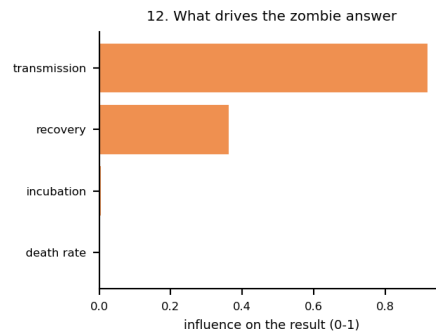




The assumed pathogen sits above every measured one. This is a restatement of the assumption, not a measurement.



Route A restates your own assumption. Route B borrows from three measured outbreaks and is the defensible way to reason about an unknown disease.



Transmission rate drives almost all of it; incubation and death rate barely matter. If data ever arrived, measure transmission first.

13. Role of each arm

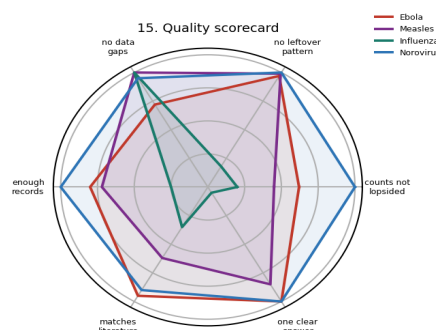
Ebola	measured	<i>carries comparison</i>
Measles	measured	<i>carries comparison</i>
Norovirus	measured	<i>carries comparison</i>
Influenza	measured	<i>method benchmark</i>
Zombie	no data	<i>hypothetical</i>

Three arms carry the comparison, one tests the method, one is a hypothetical carried without inventing data.

14. Data provenance

Ebola	RECON / Khan 1999	real
Measles	Pfeilsticker 1863 linelist	real
Influenza	BMJ 1978	real
Norovirus	O'Neill & Marks 2005	real
Zombie	assumed parameters only	no observations

No synthetic, simulated or imputed observation enters any likelihood; the zombie contributes assumptions only.



Norovirus is nearly a full hexagon; influenza collapses inward on four of six spokes.

6. Study Design and the Undocumented Pathogen

What each arm contributes, and what can honestly be claimed about the zombie

Role of each arm

Arm	Data	Role	What it licenses you to claim
Ebola	real, 139 days	carries the comparison	a spread-score estimate with an interval, comparable to published work
Measles	real, 86 days	carries the comparison	same, at the opposite end of the contagiousness scale
Norovirus	real, 492 children	carries the comparison	same, and the tightest interval of the four
Influenza	real, 14 days	method benchmark	whether the framework detects a known-hard case — it does; not a clean estimate
Zombie	none	hypothetical	a conditional statement only: IF a pathogen had these properties, THEN...

Can you best-guess a pathogen that does not exist?

Yes, in one specific sense, and no in another. The distinction is worth stating precisely because it is the difference between a defensible claim and an indefensible one.

	Route A — assumption restated	Route B — learned from the measured arms
What you do	choose values for transmission, incubation and recovery; push them through the model	let the zombie draw from the same shared prior the three measured outbreaks inform
What comes out	exactly what you put in, expressed as an outbreak	a range shaped by three real human outbreaks
Honest claim	'IF a pathogen behaved this way, THEN the outbreak would look like this'	'GIVEN three human outbreaks we measured, a fourth unmeasured one would plausibly fall here'
Not a claim	'the zombie spread score is 3.03' — that is your assumption, not a finding	any statement about this specific pathogen; it is a statement about human pathogens in general
Where it appears	figure x10	figure x11

THE SHORT ANSWER — the framework does not create information about an undocumented disease. What it does is make your assumptions explicit, show exactly what they imply, reveal which of them actually matters (transmission rate, overwhelmingly — see figure x12), and offer a second route where three measured outbreaks inform the fourth. Route B is the defensible answer to 'can this handle unknown diseases'. Route A is only a restatement, and should be labelled as such.

No synthetic data anywhere

Arm	Source of every observation	Synthetic / imputed / simulated?
Ebola	RECON outbreaks; Khan et al. (1999)	none
Measles	Pfeilsticker (1863) linelist via R surveillance package	none
Influenza	BMJ 1978;1(6112):587	none
Norovirus	O'Neill & Marks (2005)	none
Zombie	no observations exist and none were created	n/a — assumptions only

the zombie arm contributes zero observations to any likelihood. Its parameters are declared assumptions, held in a separate file, and never mixed with measured data.